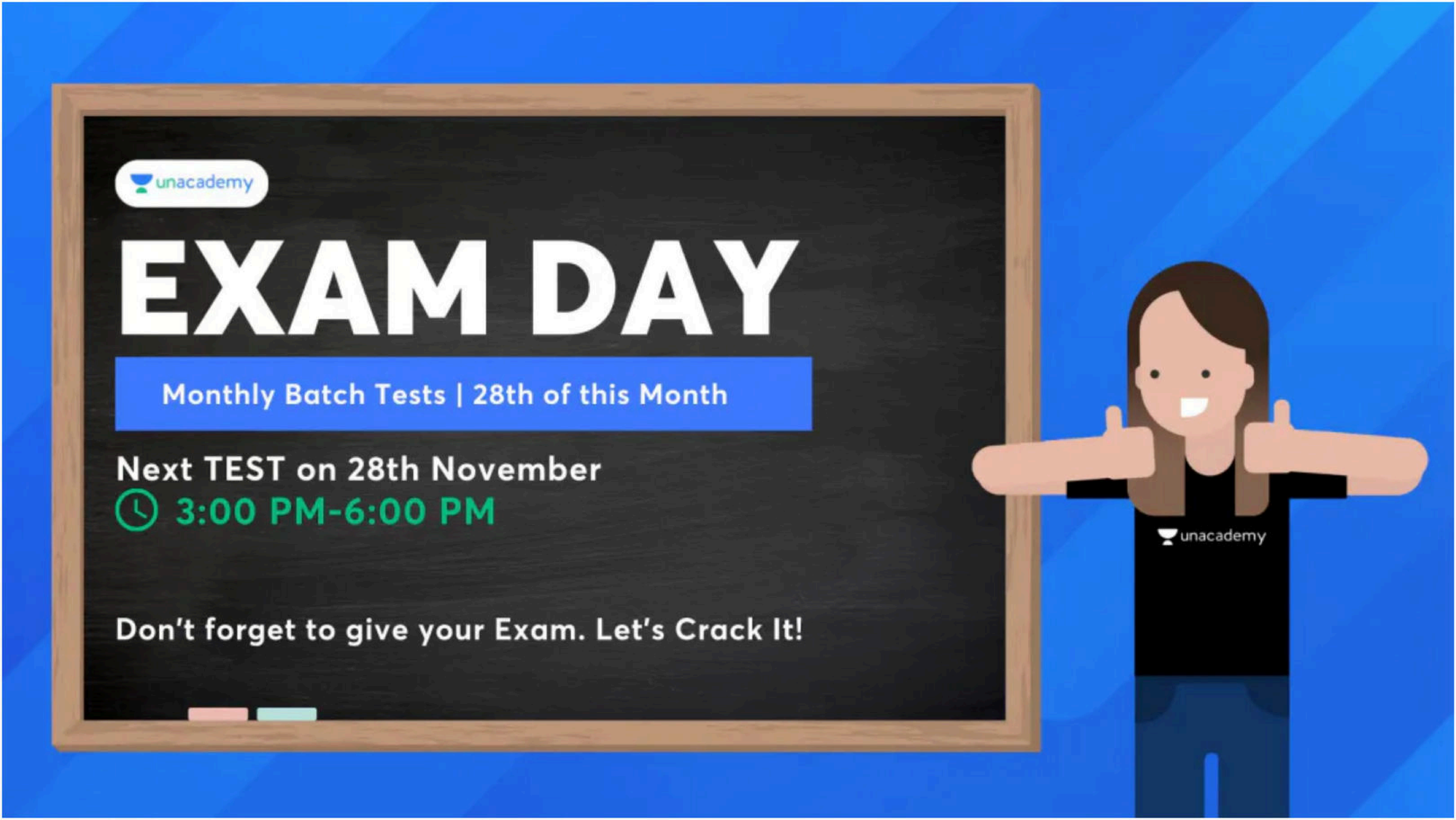
Miscellaneous Problems in Vector & 3D Geometry and Doubt Clearing Session

Comprehensive Course on Vector & 3D Geometry

Alok Kumar • Lesson 18 • Nov 27, 2021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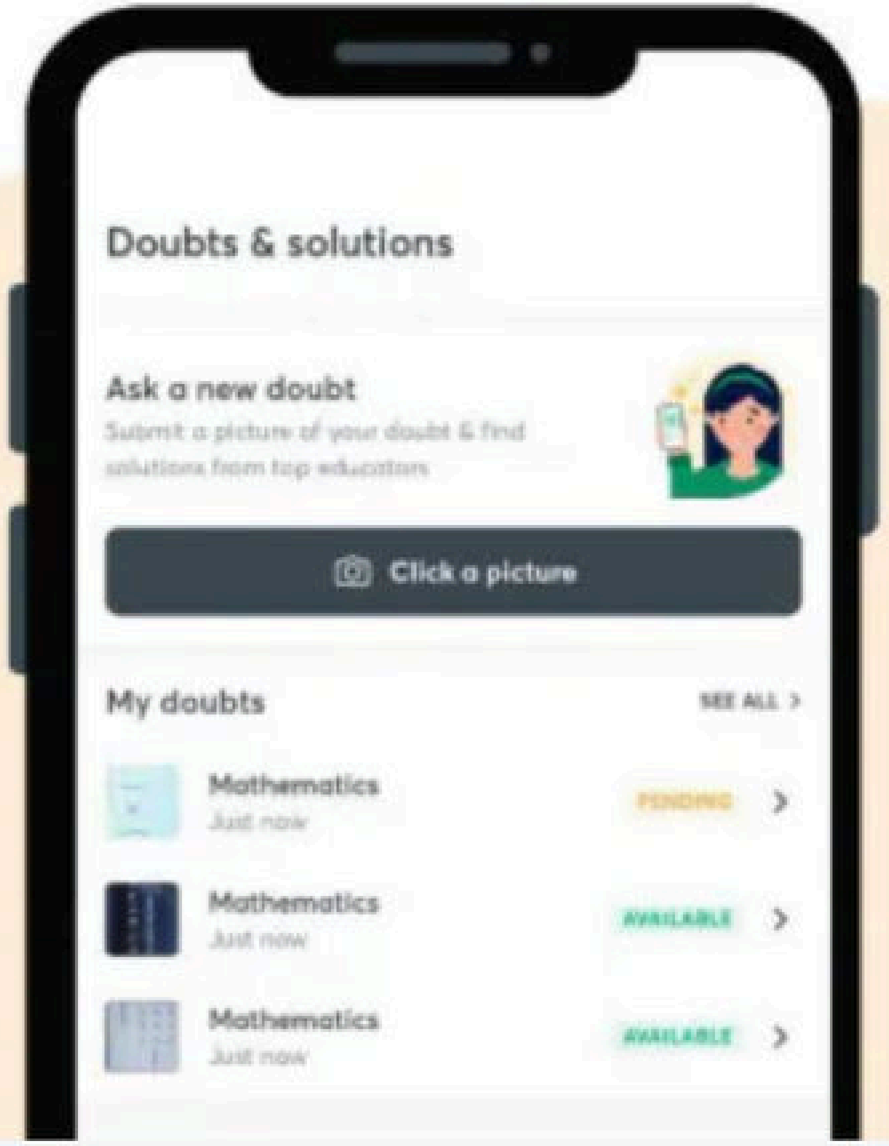

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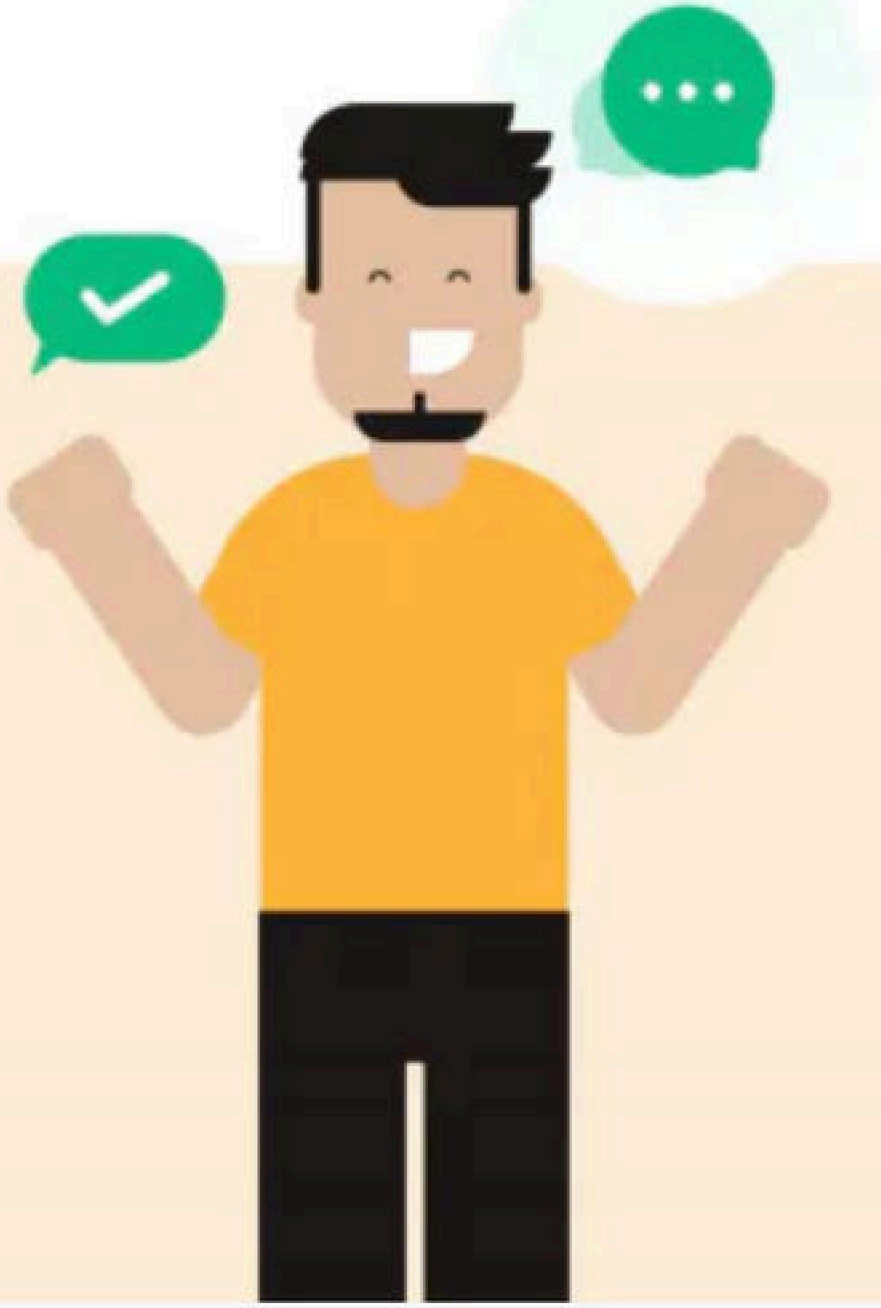

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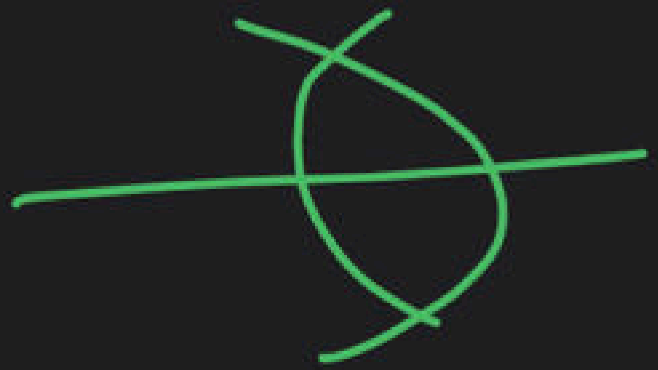
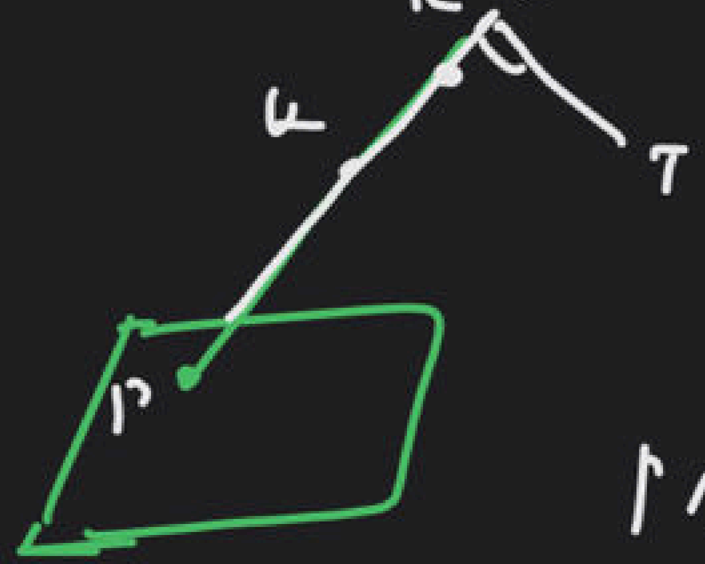
V & 3D

- DCA DR
- Pl
- line – 3D
- Tetrahedron



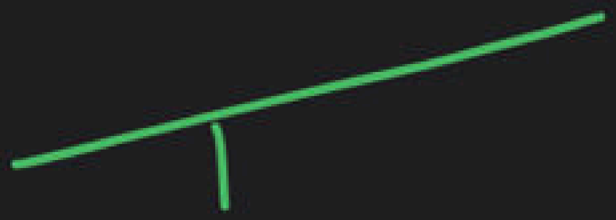
$y \overset{?}{=} a(a(x+c))$ $\frac{8}{3}\sqrt{ab}$

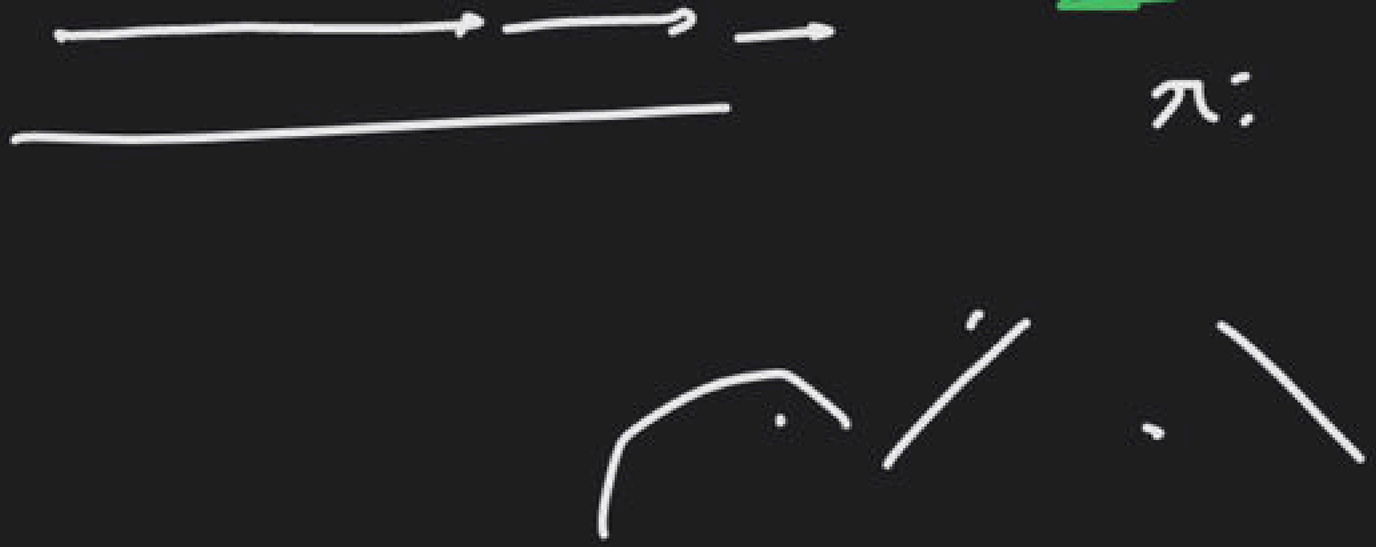
$y^2 = -4b(x+b)$ $b \to 1$



r =

π:





YT VECTORS & 3D GEOMETRY

1. If for $a > 0$, the feet of perpendiculars from the points $A(a,-2a,3)$ and $B(0,4,5)$ on the plane $lx + my + nz = 0$ are points $C(0,-a,-1)$ and D respectively, then the length of the line segment CD is equal to

   A) $\sqrt{41}$ B) $\sqrt{66}$ C) $\sqrt{55}$ D) $\sqrt{31}$

2. A variable plane passes through a fixed point (3, 2, 1) and meets x, y and z axes at A,B and C respectively. A plane is drawn parallel to yz-plane through A, a second plane is drawn parallel zx–plane through B and a third plane is drawn parallel to zy-plane through C. Then, the locus of the point of intersection of these three planes is

   A) $\frac{3}{x}+\frac{2}{y}+\frac{1}{z}=1$ B) $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{11}{6}$ C) $x + y + z = 6$ D) $\frac{x}{3}+\frac{y}{2}+z=1$

3. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{c}=\hat{j}-\hat{k}$ and a vector $\vec{b}$ be such that $\vec{a}\times\vec{b}=\vec{c}$ and $\vec{a}\,.\,\vec{b}=3$. Then, $|\vec{b}|$ is equal to

   A) $\frac{11}{3}$ B) $\frac{11}{\sqrt{3}}$ C) $\sqrt{\frac{11}{3}}$ D) $\frac{\sqrt{11}}{3}$

4. The length of the projection of the line segment joining the points (5, – 1, 4) and (4, – 1, 3) on the plane $x + y + z = 7$ is

   A) $\frac{2}{\sqrt{3}}$ B) $\frac{2}{3}$ C) $\frac{1}{3}$ D) $\sqrt{\frac{2}{3}}$

5. Let $\vec{u}$ be a vector coplanar with the vectors $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{j}+\hat{k}$. If $\vec{u}$ is perpendicular to $\vec{a}$ and $\vec{u}.\vec{b}=24, |\vec{u}|^2$ is equal to

   A) 336 B) 315 C) 256 D) 84

6. The vector equation of the plane passing through the line of intersection of the planes $x + y + z = 1$ and $2x+3y+4z=5$ which is perpendicular to the plane $x-y+z=0$ is

   A) $\vec{r}\times(\hat{i}-\hat{k})+2=0$ B) $\vec{r}.(\hat{i}-\hat{k})-2=0$

   C) $\vec{r}.(\hat{i}-\hat{k})+2=0$ D) $\vec{r}\times(\hat{i}-\hat{k})-2=0$

7. Let $a, b, c \in \mathbb{R}$ be such that $a^2+b^2+c^2=1$. If

$a\cos\theta = b\cos\left(\theta+\frac{2\pi}{3}\right) = c\cos\left(\theta+\frac{4\pi}{3}\right) = \lambda$, where $\theta = \frac{\pi}{9}$, the angle between the vectors $a\hat{i}+b\hat{j}+c\hat{k}$ and $b\hat{i}+c\hat{j}+a\hat{k}$ is

A) $\frac{\pi}{2}$ B) 0 C) $\frac{\pi}{9}$ D) $\frac{2\pi}{3}$

8. Let x_0 be the point of local maxima of $f(x) = \vec{a}.(\vec{b}\times\vec{c})$, where

$\vec{a} = x\hat{i}-2\hat{j}+3\hat{k}, \vec{b} = -2\hat{i}+x\hat{j}-\hat{k}$ and $\vec{c} = 7\hat{i}-2\hat{j}+x\hat{k}$. Then, the value of $\vec{a}.\vec{b}+\vec{b}.\vec{c}+\vec{c}.\vec{a}$ at $x = x_0$ is

A) – 30 B) 14 C) – 4 D) – 22

9. If the equation of a plane P, passing through the intersection of the planes $x+4y-z+7=0$ and $3x+y+5z=8$ is $ax+by+6z=15$ for some $a, b \in \mathbb{R}$, then the distance of the point (3, 2, – 1) from the plane P is _________ .

10. The shortest distance between the line $\frac{x-1}{0} = \frac{y+1}{-1} = \frac{z}{1}$ and line of intersection of planes $x+y+z+1=0$ and $2x-y+z+3=0$ is

A) $\frac{1}{2}$ units B) 1 units C) $\frac{1}{\sqrt{2}}$ units D) $\frac{1}{\sqrt{3}}$ units

11. The vertices B and C of ΔABC lie on the line $\frac{x+2}{3} = \frac{y-1}{0} = \frac{z}{4}$ such that $BC = 5$ units. Then, the area (in sq.units) of this triangle is (the coordinates of point A are (1, - 1, 2))

A) $2\sqrt{34}$ B) $\sqrt{34}$ C) 6 D) $5\sqrt{17}$

12. In ΔABC, right-angled at vertex A, if the position vectors of A, B and C are respectively $3\hat{i}+\hat{j}-\hat{k}, -\hat{i}+3\hat{j}+p\hat{k}$ and $5\hat{i}+q\hat{j}-4\hat{k}$, then point ($p$, q) lies on a line

A) is parallel to the y–axis.

B) makes an acute angle with the positive direction of the x-axis

C) is parallel to the x–axis

D) makes an obtuse angle with the position direction of the x–axis.

13. If $\vec{u}, \vec{v}, \vec{w}$ are non-coplanar vectors and p, q are real numbers, then the equality $\left[3\vec{u}, p\vec{v}, p\vec{\omega}\right] - \left[p\vec{v}, \vec{\omega}, q\vec{u}\right] - \left[2\vec{\omega}, q\vec{v}, q\vec{u}\right] = 0$ holds for:

    A) exactly two values of (p,q)

    B) more than two but not all values of (p,q)

    C) all values of (p,q)

    D) exactly one value of (p,q)

14. Let $ABCD$ be a parallelogram such that $\overrightarrow{AB} = \vec{q}, \overrightarrow{AD} = \vec{p}$ and $\angle BAD$ be an acute angle. If $\vec{r}$ is the vector that coincide with the altitude directed from the vertex B to the side AD, then $\vec{r}$ is given by:

    A) $\vec{r} = 3\vec{q} - \dfrac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$ B) $\vec{r} = -\vec{q} + \dfrac{(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

    C) $\vec{r} = \vec{q} - \dfrac{(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$ D) $\vec{r} = -3\vec{q} - \dfrac{3(\vec{p}.\vec{q})}{(\vec{p}.\vec{p})}\vec{p}$

15. The equation of the plane containing the line $2x - 5y + z = 3$; $x + y + 4z = 5$, and parallel to the plane, $x + 3y + 6z = 1$, is :

    A) $x + 3y + 6z = 7$ B) $2x + 6y + 12z = -13$

    C) $2x + 6y + 12z = 13$ D) $x + 3y + 6z = -7$

16. The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along the line $x = y = z$ is:

    A) $\dfrac{10}{\sqrt{3}}$ B) $\dfrac{20}{3}$ C) $3\sqrt{10}$ D) $10\sqrt{3}$

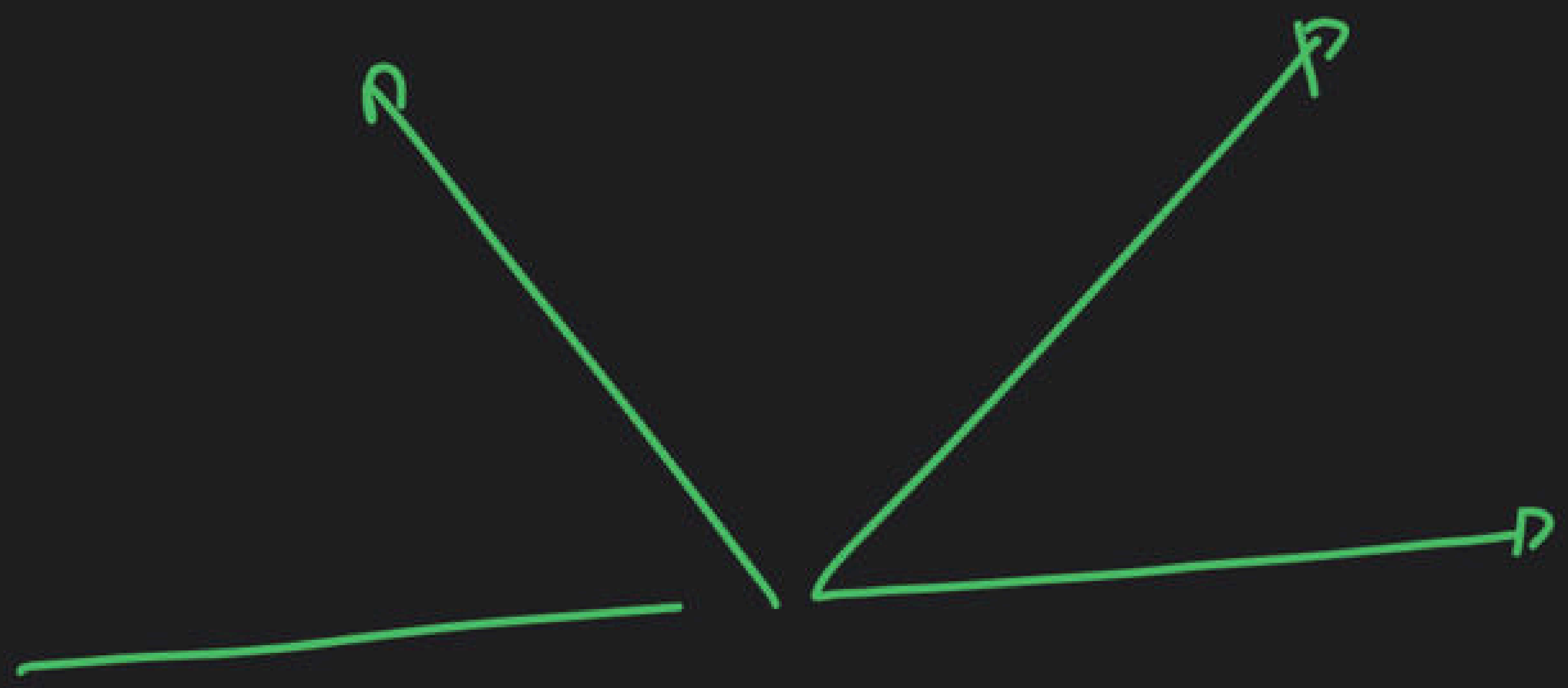
17. Let $\vec{a} = \alpha\hat{i} + 2\hat{j} + \beta\hat{k}$ lies in the plane of $\vec{b}$ and $\vec{c}$, where $\vec{b} = \hat{i} + \hat{j}$ and $\vec{c} = \hat{i} - \hat{j} + 4\hat{k}$. If $\vec{a}$ bisects the angle between $\vec{b}$ and $\vec{c}$, then

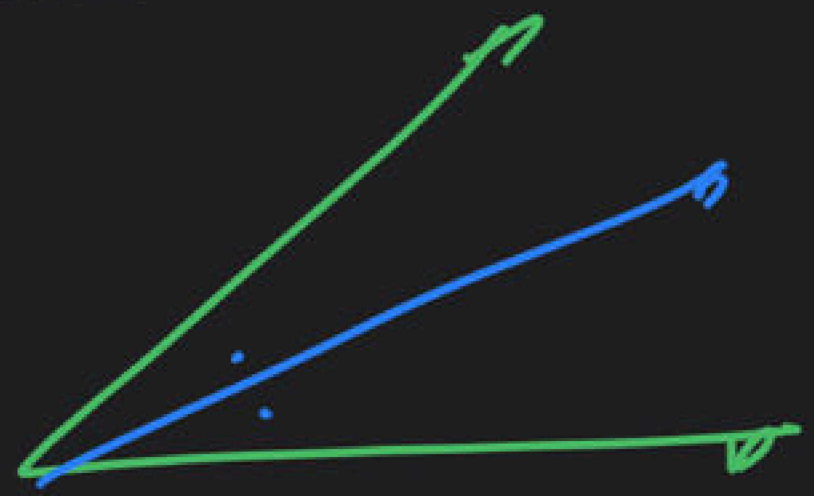
    A) $\vec{a}.\hat{k} + 2 = 0$ B) $\vec{a}.\hat{k} + 4 = 0$ C) $\vec{a}.\hat{k} - 2 = 0$ D) $\vec{a}.\hat{k} + 5 = 0$

18. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = 0$ $\lambda = \vec{a}\,.\,\vec{b} + \vec{b}\,.\,\vec{c} + \vec{c}\,.\,\vec{a}$ and $\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$, then $(\lambda, d) =$

    A) $\left(\dfrac{3}{2}, 3\vec{a} \times \vec{b}\right)$ B) $\left(-\dfrac{3}{a}, 3\vec{a} \times \vec{c}\right)$ C) $\left(-\dfrac{3}{2}, 3\vec{a} \times \vec{b}\right)$ D) $\left(\dfrac{3}{2}, 3\vec{a} \times \vec{c}\right)$

$(\alpha \;\; 2 \;\; \beta) = \lambda(1, 2, -2)$





$\alpha = \lambda$

$2 = 2\lambda$

$\beta = -2\lambda$

$\lambda = 1$

$(\alpha \;\; 2 \;\; \beta) = \lambda(2, 1, 2)$

$\alpha = 1$

$\beta = -2$

$\alpha = 2k$

$2 = k$

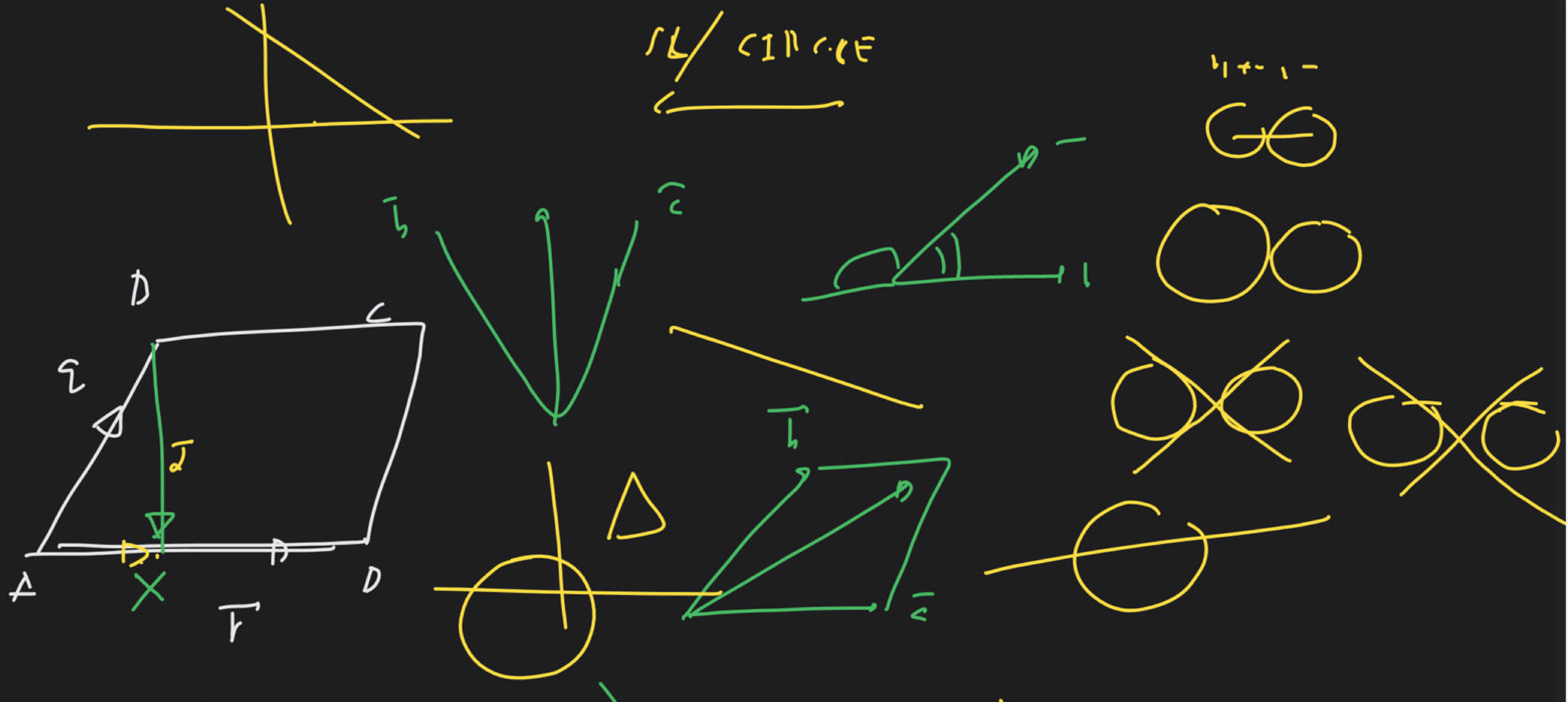
$\beta = 2k$

$\alpha = 4$

$\beta = 4$

$$\begin{vmatrix} \alpha & 2 & \beta \\ 1 & 1 & 0 \\ 1 & -1 & 4 \end{vmatrix} = 0 \Rightarrow$$

$2\alpha - \beta = 4$

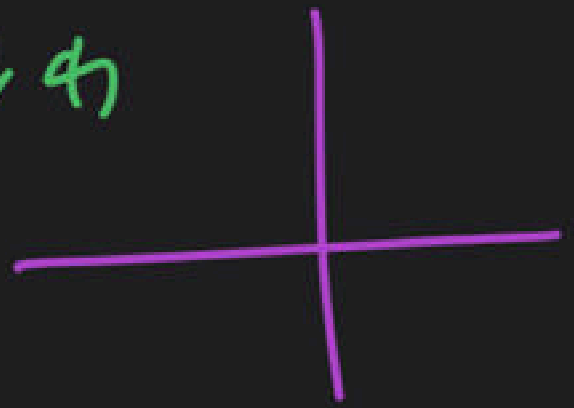


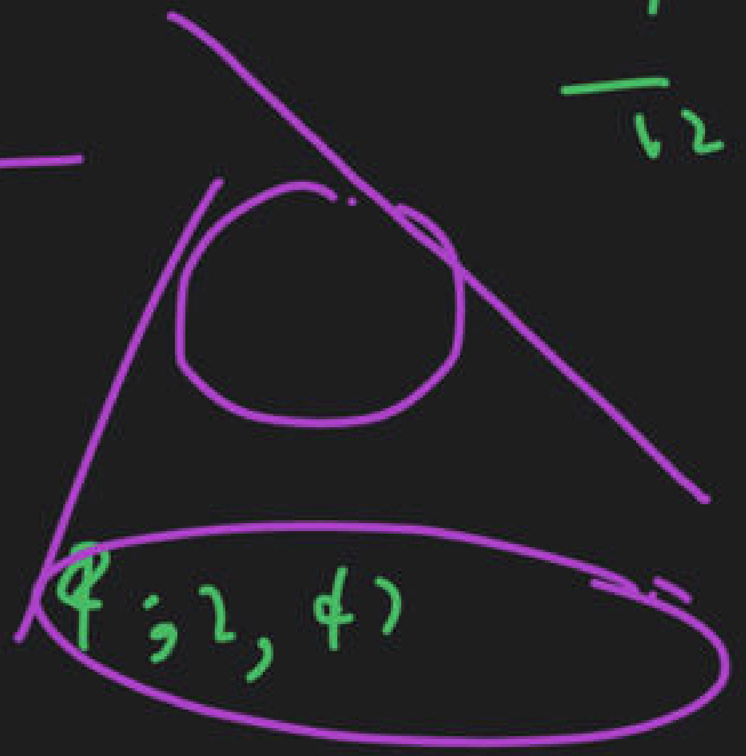
$$\frac{\vec{e}\cdot\vec{r}}{|r|^2}\vec{r} + (-\vec{a}) = \vec{e}$$

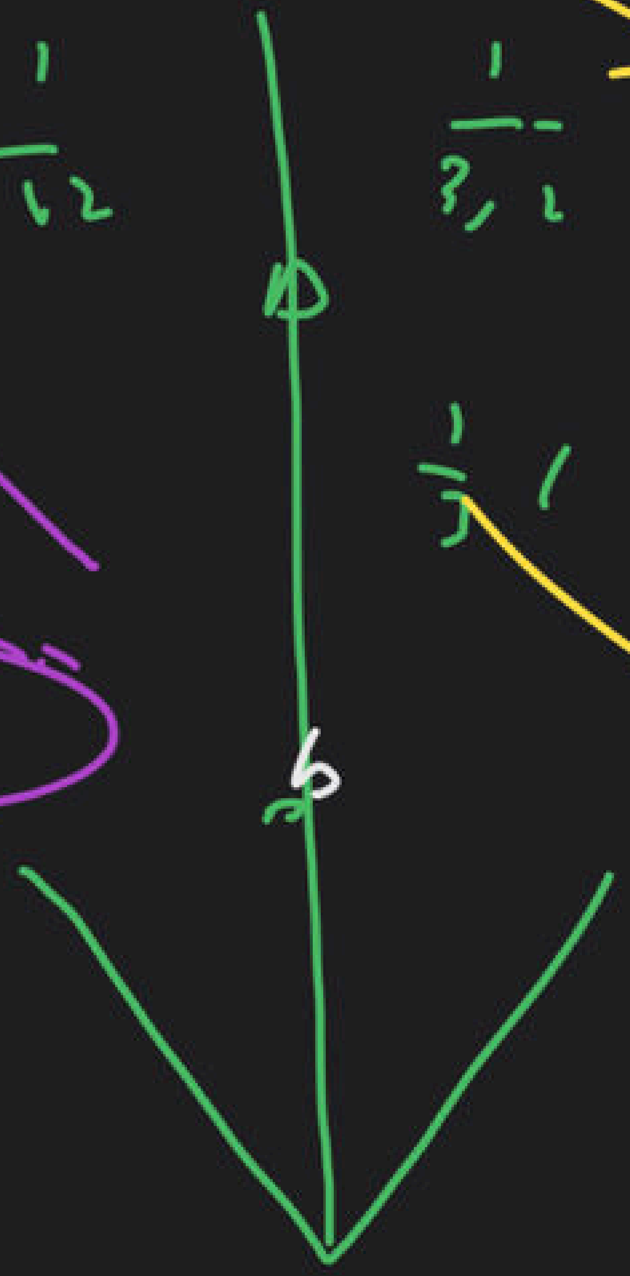
$b; c$

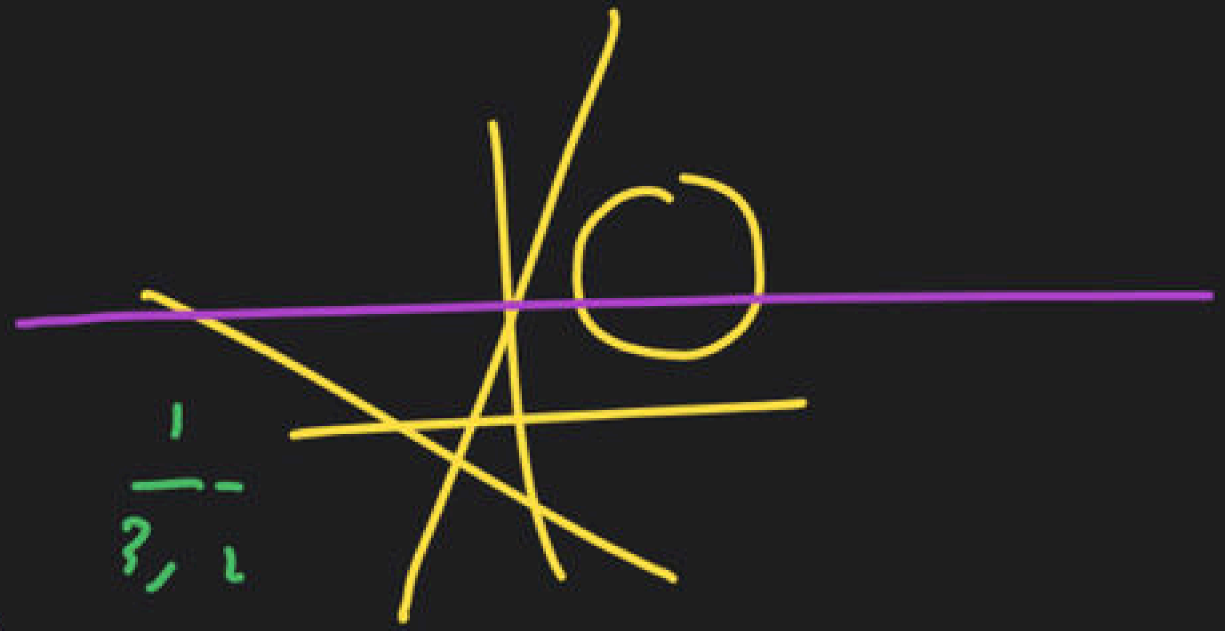
$3(1,1,\mu 0) - (1,-1,4)$

$+$









$\frac{1}{\sqrt{2}}$ $\frac{1}{3,2}$

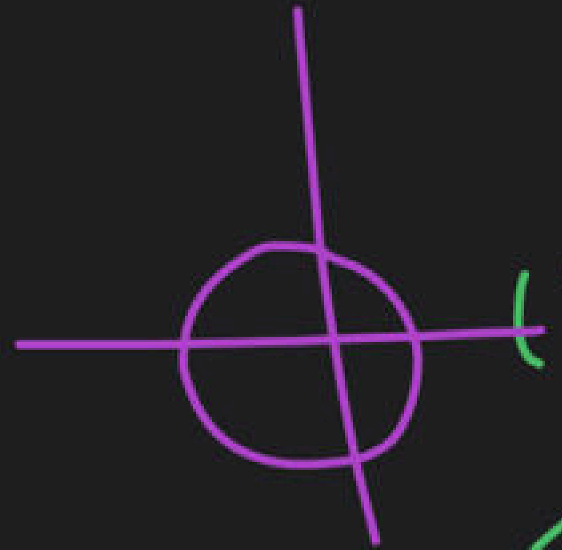
$\frac{1}{3}(\ ,$

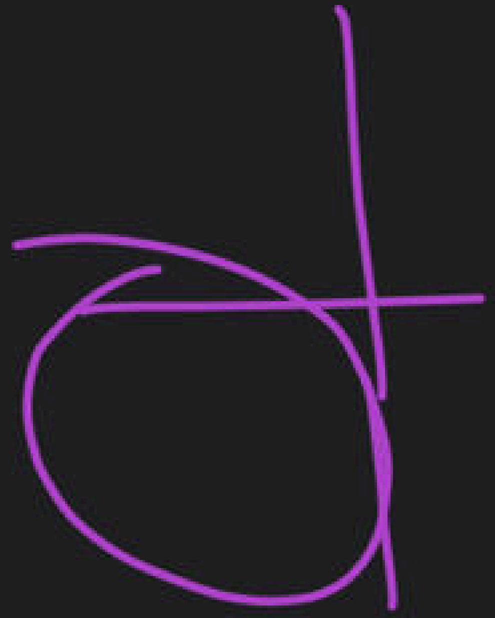
$(2,4,-4)$

$(4;2,4)$

$\langle 1,2,-2)$

$\langle 2,1,2)$





$\bar{a} = (1,2,-2)$

$\bar{k} =$

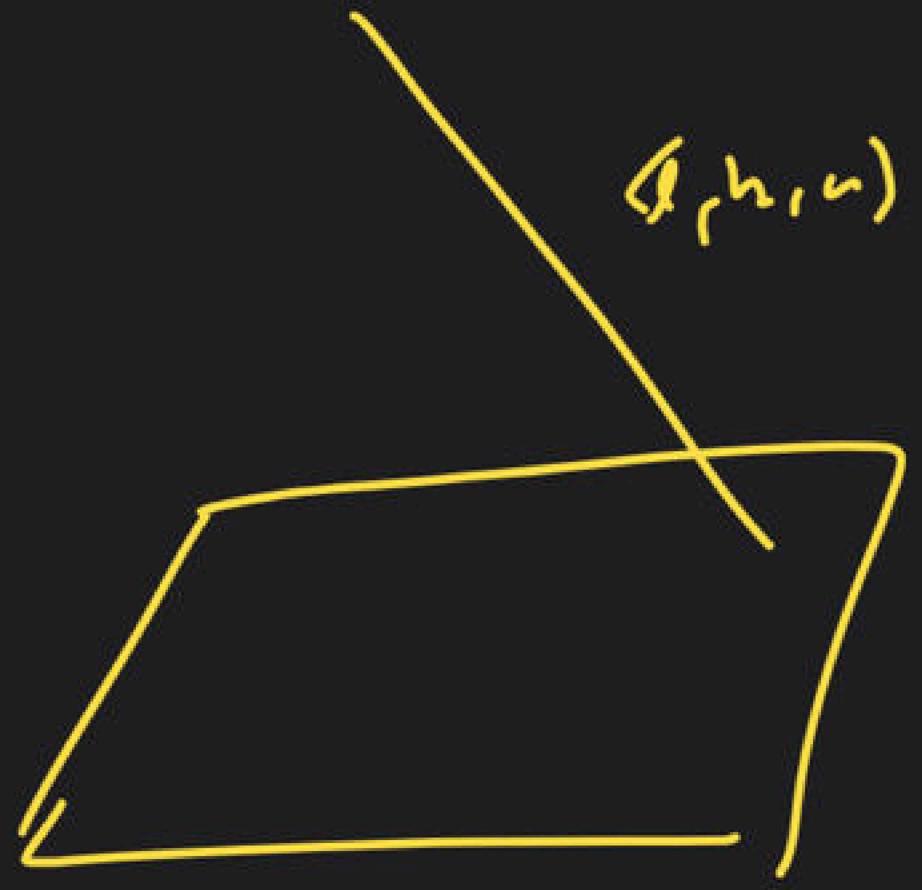
$\bar{a}\cdot\bar{i} = -2$

$\vec{a} = \lambda(\hat{b}+\hat{c})$

$\bar{a} = \lambda(1,2,-2)$ $\bar{c} = \mu(2,1,2)$

$\boxed{\bar{r} = \alpha \cdot H(2) + \beta\hat{i}}$

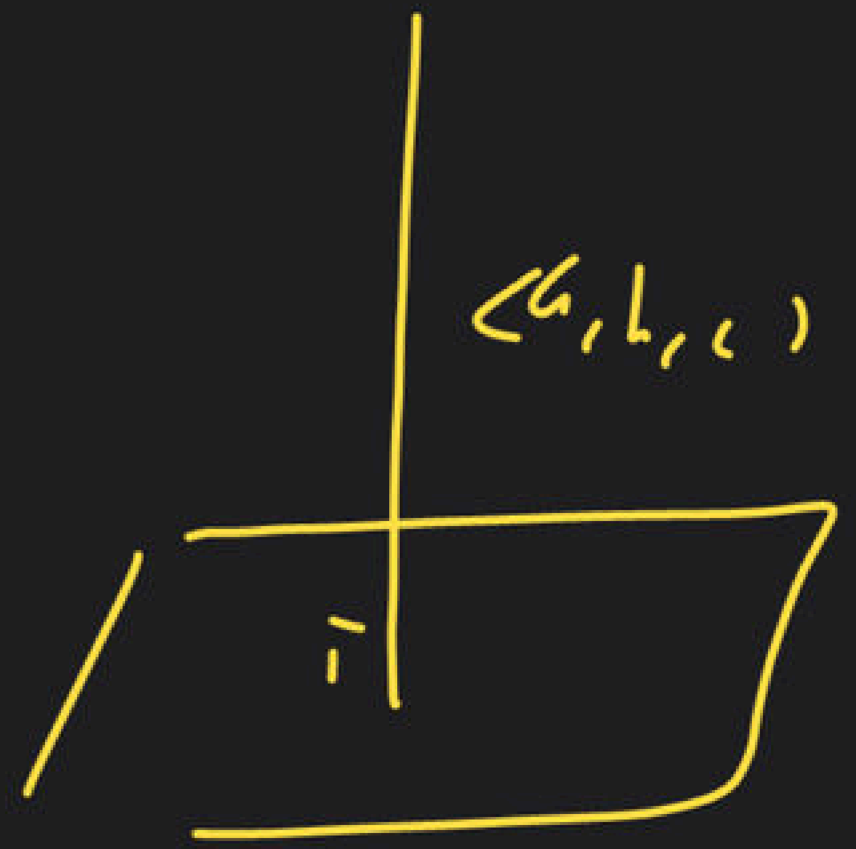
$x = 2$ $x = 2$



$ax+by=c$

(a,b,c)

$ax+by+cz$



c^2+b^2

$+\sqrt{3}$

19. If $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c}$ is non-zero vector and $\vec{b} \times \vec{c} = \vec{b} \times \vec{a}$, $\vec{a}.\vec{c} = 0$, then $\vec{b}.\vec{c}$ is equal to

A) $\frac{1}{2}$ B) $-\frac{1}{3}$ C) $-\frac{1}{2}$ D) $\frac{1}{3}$

20. Let the distance between the plane passing through lines

$$\frac{x+1}{2} = \frac{y-3}{3} = \frac{z+1}{8}$$

$\frac{x+3}{2} = \frac{y+2}{1} = \frac{z-1}{\lambda}$ and the plane $23x - 10y - 2z + 48 = 0$ is $\frac{k}{\sqrt{633}}$, then the value of k is

A) 4 B) 3 C) 2 D) 1

Lakshay:

$$\pi_1: \begin{cases} \dfrac{x-4}{2} = \dfrac{y-3}{4} = \dfrac{z+1}{8} \\[2ex] \dfrac{x+3}{2} = \dfrac{y+2}{1} = \dfrac{z-1}{\lambda} \end{cases}$$

2, 3, 8

2, 1, λ

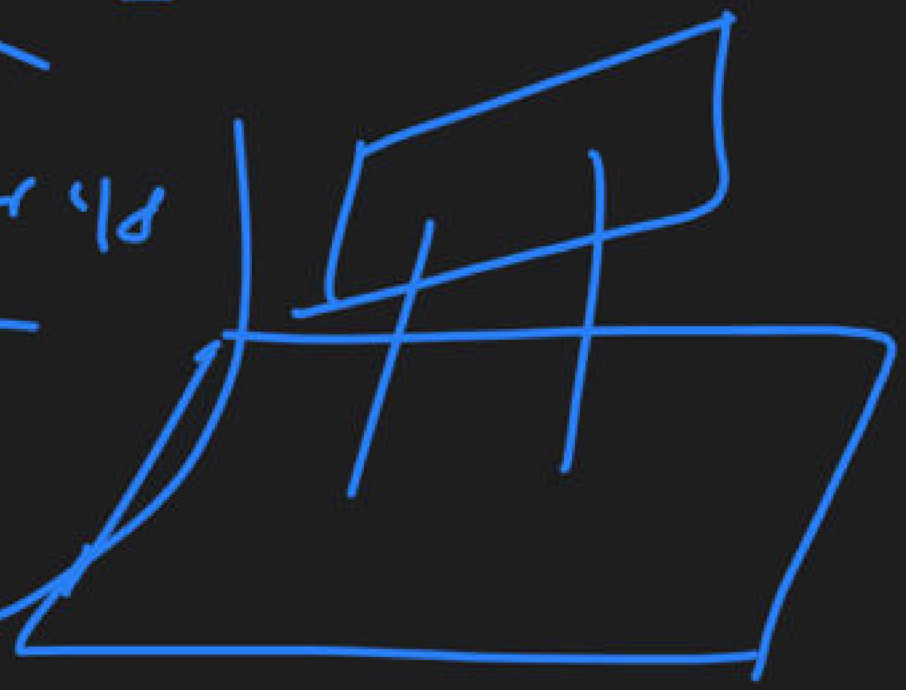
$$\begin{vmatrix} 2 & 3 & 8 \\ 2 & 1 & \lambda \\ 2 & -5 & 2 \end{vmatrix} = \ldots$$

$(-1, 3, -1)$

$$\times \quad \begin{matrix} \langle 2, 3, 8 \rangle \\ \langle 2, -5, 2 \rangle \end{matrix}$$

$\pi_2 \quad 23x - 1y - 2z + 48 = 0$

$$\frac{-23 - 3 + 1 + 48}{\sqrt{23^2 \ldots}}$$



d, ...

$\boxed{23 - 1 \ldots = -}$